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UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION

Midcontinent Independent System Operator, Inc.	**)**	**Docket No. ER26-___-000**

PREPARED DIRECT TESTIMONY OF
MARC KEYSER
DIRECTOR, CENTRAL REGION—EXTERNAL AFFAIRS

I. PROFESSIONAL BACKGROUND AND QUALIFICATIONS

Q. PLEASE STATE YOUR NAME, BUSINESS ADDRESS, AND RELATIONSHIP TO THE MIDCONTINENT INDEPENDENT SYSTEM OPERATOR, INC.

A. My name is Marc Keyser. I am the Director, Central Region—External Affairs for the Midcontinent Independent System Operator, Inc. ("MISO"). My business address is 720 City Center Drive, Carmel, IN 46032.

Q. PLEASE BRIEFLY DESCRIBE YOUR EDUCATION AND PROFESSIONAL BACKGROUND.

A. I hold a Bachelor of Science and Engineering Degree in Nuclear Engineering, a Bachelor of Science and Engineering Degree in Engineering Physics, and a Master of Science and Engineering Degree in Nuclear Engineering – Plasma Physics. I have been employed by MISO since February, 2007 in a variety of different roles. I spent six years in a market design and market / operational engineering role. As MISO worked to launch its Energy and Ancillary Services Markets, I led the design of the Automatic Generation Control System. Since that time, I have held leadership roles in Client and Stakeholder Relations, Corporate Strategy, and Program Management. Prior to joining MISO, I worked from 2002-2007 in various system operation roles for Consumers Energy, an investor-owned utility in Michigan.

Q. PLEASE DESCRIBE YOUR CURRENT ROLE AND RESPONSIBILITIES.

A. In my current role, I provide advisory services across MISO business areas and manage relationships with key MISO stakeholders across the central region of the MISO footprint. Given my length and breadth of experience, I was chosen to act as the MISO liaison to the newly-formed Large Load Working Group ("LLWG") in March 2026. As the LLWG

liaison, I coordinate MISO's planned activities with respect to our large load stakeholder efforts. Since October 2025, I have also led MISO's effort to create and formalize the proposal accompanying my testimony. As part of this responsibility, I have closely examined existing and proposed co-location arrangements in the MISO region and the rule sets that govern them.

Q. HAVE YOU PREVIOUSLY SUBMITTED TESTIMONY IN A REGULATORY PROCEEDING?

A. No**.** This is the first testimony that I have provided in any regulatory proceeding**.**

II. PURPOSE OF TESTIMONY

Q. WHAT IS THE PURPOSE OF YOUR TESTIMONY?

A. My testimony supports two related items. First, my testimony supports the proposed Tariff revisions submitted in this docket to implement Zero Injection Generator Interconnection Service ("ZG Interconnection Service") and a Zero Injection Generator Interconnection Agreement ("ZGIA") to govern and memorialize the terms of this service. Second, my testimony also explains how the ZGIA addresses certain findings and directives in the Commission's Show Cause Order related to large load integration issued to MISO in Docket No. EL26-70-000 on June 18, 2026 ("Show Cause Order"). This includes, but is not limited to, how the ZGIA process aims to provide Speed to Power by increasing the pace of generating capacity additions in the MISO region and to provide a mechanism to better evaluate the reliability impacts of co-location and similar arrangements. In providing this testimony, capitalized terms not otherwise defined herein have the meeting set forth in MISO's Tariff.

Q. WHY IS THE ZGIA CONSTRUCT NEEDED?

A. ZGIA implementation is needed to: (1) provide MISO with visibility into generating resources that plan to connect to the Transmission System but not inject onto the Transmission System; and (2) the ZGIA will allow MISO to establish a framework to provide the necessary studies to ensure reliability of the Transmission System, without subjecting generators that are not expected to have significant system impacts to the more protracted interconnection queue. The ZGIA construct also provides a benefit to the interconnection queue, because it provides a parallel path for differently situated generators to be studied for interconnection service.

Q. PLEASE EXPLAIN WHY THIS PROPOSAL IS TIME SENSITIVE.

A. As discussed in the Show Cause Order, the Commission has taken numerous actions in the past year to address unprecedented increases in requests for transmission service to serve large loads and the growing interest in the co-location of loads with generating facilities (*i.e.,* co-location agreements). These actions, coupled with the Commission ANOPR addressing the need to take additional actions related to the addition of large and co-located loads, highlight the urgent need to implement solutions to address large load integration and co-location arrangements in the MISO region.

MISO agrees that it is experiencing an unprecedented surge in large load additions, primarily from data centers. In recognition of the urgent need to address speed to reliable power and associated network upgrades, MISO established the Expedited Resource Addition Study ("ERAS") process to accelerate the generator interconnection process and improved MTEP's Expedited Project Review ("EPR") process to quickly approve transmission network upgrades needed to integrate large loads. While existing processes have effectively integrated new large loads, MISO needs to continue to evolve its approach

and develop long-term solutions to maintain reliability while enabling accelerating load growth.

This accelerating load growth poses an increasing reliability challenge to the grid, and has been discussed, documented and highlighted in various MISO and non-MISO reports.[1] Finally, large load can have different enablement needs and pose operational challenges and opportunities that differ from traditional load additions. This filing is urgently needed to provide an enablement path directly responsive to the specific needs of co-located generation and large loads. As MISO continues to receive such requests, having a ZGIA process available will improve the speed, efficiency, and reliability of such integrations.

III. THE CURRENT STATE OF THE MISO QUEUE AND THE NEED FOR ZG INTERCONNECTION SERVICE

Q. PLEASE DESCRIBE THE EFFECTIVENESS OF MISO'S INTERCONNECTION QUEUE PROCESS FOR ADDRESSING THE NEEDS OF LARGE LOADS.

A. MISO's Interconnection Queue, known as the Definitive Planning Phase ("DPP"), provides a valuable tool that processes a significant volume of generation requests each study cycle. The DPP processes most Interconnection Requests for new Generating Facilities and Energy storage Resources. It will continue to be the primary tool for meeting most load growth needs, including some large loads.

[1] 1.EIA - https://www.eia.gov/todayinenergy/detail.php?id=67704 , 2.Berkeley Labs report - https://bies.lbl.gov/news/berkeley-lab-report-evaluates-increase-electricity-demand-data-centers , 3. 2026 OMS Survey results - https://cdn.misoenergy.org/20260604%20OMS-MISO%20Survey%20Results%20Workshop%20Presentation761711.pdf, 4, 2026 PRA Results - https://cdn.misoenergy.org/2026%20PRA%20Results%20Posting%2020260522%20-%20Corrections754715.pdf.

However, the number of generator interconnection requests in the MISO queue continues to be extraordinary and unprecedented. As of July 2026, MISO's generator interconnection queue contains 802 active Interconnection Requests comprising 167 GW of proposed new generator interconnection capacity. This considerable volume of applications is spread over all five of MISO's study regions and includes queue cycles going back to 2021. MISO has engaged in significant reform efforts to address the speed of processing the queue. For example, since MISO's last major queue reform in 2017, MISO has made numerous improvements to its queue rules. In 2019, MISO made a filing with the Commission to strengthen certain milestone and site control requirements. In 2021, MISO filed improvements to its Option to Build, harm test calculation, and provisional GIA requirements as well as filings to classify fuel type change requests made in the DPP process as a Material Modification and allow Interconnection Customers to request Surplus Interconnection Service earlier in the interconnection study process. In 2022, MISO filed an innovative proposal to reduce the GIP timeline by offering enhanced flexibility and an optional path for Interconnection Customers to execute GIAs in under one year. In 2023, during the pendency of the Order No. 2023 process, MISO further strengthened its DPP process and worked to discourage speculative Interconnection Requests through adoption of an Automatic Withdrawal Penalty and further refinements of its site control requirements. In addition, MISO worked with its RTO neighbors, PJM Interconnection, LLC ("PJM") and Southwest Power Pool, Inc. ("SPP"), to improve affected system studies procedures and requirements that are contained in MISO's Joint Operating Agreements. In addition to these efforts, MISO and SPP developed and filed the Joint Targeted Interconnection Queue Study ("JTIQ") process, which has sped the queue up considerably

by providing an alternative to the backlogged affected systems study process with SPP while providing both more efficient solutions to the needs of multiple cycles of Interconnection Customers and greater up-front cost certainty regarding potentially significant affected systems costs. In 2024 and 2025, MISO developed and filed an innovative ERAS process as a temporary solution to address urgent near-term resource adequacy and reliability concerns due to load growth, generation retirement, and delays in the interconnection process. Beyond these filed processes, MISO has worked to develop internal process and software improvements, such as integrated modeling software Suite of Unified Grid Analyses with Renewables ("SUGAR"), for Phase 1 DPP studies with plans to integrate in Phase 2 and 3. The SUGAR software leverages powerful circuit simulation methods developed to study billion-transistor electronic systems. SUGAR reduced DPP phase 1 cycle time significantly from the status quo as evidenced in the DPP 2022, 2023 and 2025 cycles.

While these reforms are producing—and will continue to produce—significant process efficiencies and allow MISO to shorten the study time needed, it would not be feasible or desirable to truncate the DPP to the degree needed to meet the short lead times demanded by data centers and other large loads to meet their need by dates—some of which seek to go online within a year—without compromising the utility for the DPP for all other types of Interconnection Requests.

The DPP enables system reliability by studying new generation sources for technical and operational impacts on the Transmission System and identifying and assigning the costs for the necessary upgrades required to connect the new resources safely and reliably. However, in limited circumstances involving generation serving load at the

same substation or behind the same Point of Interconnection where the proposed configuration presents a minimal likelihood of impacting the Transmission System, a separate, expedited study process and corresponding form of Interconnection Service is feasible and needed to meet the frequently short lead times required by certain loads, including large loads associated with data centers.

Q. PLEASE DESCRIBE THE RATIONALE FOR MISO'S PROPOSAL TO ESTABLISH A NEW FORM OF INTERCONNECTION SERVICE.

A. MISO is proposing a new form of interconnection service because the existing service types are studied in a manner that is inherently broader, because they study more significant far reaching impacts to the Transmission System. This type of study is unnecessarily time-consuming and expensive for Interconnection Customers who seek to serve load at the same substation.

Some Interconnection Customers—particularly those serving large loads— seek to install generation to meet the needs of adjacent large loads, do not require access to the transmission system and, in fact, may never intend to inject onto the MISO Transmission System or, if they do plan on injecting, it is only to move energy to an adjacent bay of the same substation. Under the current Tariff, these Interconnection Customers are required to pursue at a minimum ERIS, which grants "as available" access to the full MISO Transmission System. As a result, the ERIS studies must evaluate potential impacts to the entire MISO Transmission System. These studies must be robust, lengthy, and often performed on groups of Interconnection Requests because they can affect the same constraints. Such a study process is excessive for customers whose electrical impacts are confined to a single substation or adjacent load.

It would be unnecessary and burdensome to require Interconnection Customers who do not plan to inject onto the Transmission System to proceed through the DPP due to their limited needs. It is unnecessary for Interconnection Customers willing to commit to highly limited injection because the ERIS study is configured for broad "as available" system access and the Network Upgrades needed to allow such service would not be required by the limited Interconnection Customer. Requiring Interconnection Customers participating in these limited co-location arrangements to obtain ERIS would require MISO either to model the impacts of these units far more narrowly than other Interconnection Customers to accurately represent their impact, or to ignore the limitations on their injection and model them as Interconnection Requests that are not as limited as planned. The latter approach would inflate the costs of such co-location arrangements, and the former approach essentially adds delays to what can be a much narrower and faster study in order to maintain consistency of timing in the study group. Neither approach is efficient or likely to promote the expeditious development of generation to serve adjacent large loads. Despite the inefficiency of the DPP with respect to differently situated large loads, and the mismatch of existing interconnection service products for Interconnection Customers serving a single substation or adjacent load, some form of interconnection service is needed to: (1) address the actual impacts of limited injection and ensure the Interconnection Customer pays for any necessary upgrades associated with these limited impacts; (2) validate the sufficiency of the Interconnection Customer's proposed mechanisms to limit injection; and (3) memorialize commitments, including control equipment and operational limitations, in a binding agreement. MISO developed ZG

Interconnection Service and the resulting ZGIA to provide transparency through a more targeted and efficient study process.

Q. PLEASE DESCRIBE THE PURPOSE OF THE ZERO INJECTION GENERATOR INTERCONNECTION AGREEMENT.

A. The ZGIA construct serves a number of purposes, including: (i) providing visibility into Transmission connected generation that does not plan to inject onto the Transmission System; (ii) providing a needed framework to confirm lack of injection and enforce commitments; (iii) enabling a streamlined interconnection process for projects that do not have significant system impacts; (iv) providing a lower cost Interconnection Service that will be highly useful to large loads; and (v) ensuring that general requirements that should apply to all generators connected to the Transmission System are uniformly applied.

Q. PLEASE DESCRIBE HOW THE ZG INTERCONNECTION SERVICE REQUIREMENTS DIFFER FROM AN INTERCONNECTION REQUEST IN THE MISO QUEUE?

A. The ZG Interconnection Service requirements differ from those applicable to Interconnection Requests in the queue in several respects. First, the resource seeking ZG Interconnection Service must demonstrate a documented relationship with load connected at the same voltage side of the same substation. This ZG Associated Load must affirm its intention to be served by the ZG Resource. As ZG Interconnection Service limits the ZG Resource to serving only its designated ZG Associated Load, documenting this relationship is necessary to prevent speculative Interconnection Requests that are not tied to actual load. Second, the Interconnection Customer seeking ZG Interconnection Service, while not limited in the quantity of nameplate capacity of ZG Resource they choose to build, is

limited to requesting and being awarded Interconnection Service at a level that does not exceed the peak demand of the ZG Associated Load. This helps ensure that expected production does not exceed load and therefore helps prevent injections onto the Transmission System. Third, the ZG Interconnection Service Study must model the ZG Associated Load when studying the Interconnection Request and confirm that the proposed equipment to limit injections is sufficient. Fourth, the ZG Resource must interconnect at the same substation and voltage level as the Point(s) of Withdrawal through which it will serve the ZG Associated Load. Fifth, the ZG Interconnection Study must determine that the ZG Resource will not require Network Upgrades beyond the substation except for system protection facilities at adjacent substations. If all of these criteria are met, the ZG Interconnection study process affords an accelerated study process leading to a ZGIA within 90 days that memorializes the proposed configuration and limited injection.

Q. WHAT ARE THE KEY DISTINCTIONS BETWEEN THE STUDY PERFORMED UNDER THE DPP PROCESS AND THE STUDY CONDUCTED FOR A ZGIA?

A. The study performed under the DPP process differs from the ZGIA study in several important respects. Under the DPP, Interconnection Requests are evaluated through a three-phase, grouped study process that includes decisions points designed to identify and remove speculative projects as more information is discovered about the impact of adding a generating resource to the Transmission System. Because projects all impact the Transmission System and are studied in groups, each Interconnection Request is affected by higher-queued and same-group projects, introducing complexity and uncertainty, particularly when withdrawals occur.

The ZGIA study, in contrast, will be conducted on a serial, first-come, first-served basis, rather than in groups. The ZGIA study incorporates analyses specific to the limited-injection nature of the ZG Interconnection Service to ensure system reliability, including a required scenario where the ZG Associated Load trips offline. Due to the different nature of the interconnection of ZG Resources, the ZGIA study framework is sensible because of the minimal impacts beyond the substation and therefore can be completed on a faster timeline, in parallel to the DPP.

Q. WHY IS THE AMOUNT OF SERVICE REQUESTED AND AWARDED LIMITED TO THE PEAK DEMAND OF THE ZG ASSOCIATED LOAD AND THE NAMEPLATE CAPACITY OF THE ZG RESOURCE IS NOT LIMITED?

A. The service award limitation of ZG Interconnection Service serves to ensure that the maximum expected energy production of the ZG Resource is no greater than the maximum demand of the ZG Associated Load, and as a result, the ZG Resource does not inject more broadly onto the Transmission System. The additional nameplate capacity allows the ZG Resource to achieve sufficient accredited capacity to meet the ZG Associated Load within the MISO capacity construct, where accreditation varies by resource type. It is also permissible because nameplate capacity does not determine the level of Interconnection Service. An Interconnection Customer may not inject in excess of its Interconnection Service level.

Q. WHY DOES MISO PROPOSE TO LIMIT ZG INTERCONNECTION SERVICE TO ARRANGEMENTS WHERE THE ZG RESOURCE AND THE ZG ASSOCIATED LOAD ARE LOCATED ON THE SAME VOLTAGE SIDE OF THE SAME SUBSTATION?

A. MISO developed this requirement after extensive stakeholder discussions and considering various alternatives including both limiting ZG Interconnection Service to the same Point of Interconnection or, conversely, expanding it to allow the generation and ZG Associated Load to be separated by one or more substations. As part of these discussions, MISO received strong stakeholder opposition to broadening ZG Interconnection Service to embrace what the Show Cause Order refers to as electrically proximate load. Stakeholders, including developers, Transmission Owners, and other entities argued that this broader scope was more than what was needed to enable large loads and would undermine the ability to ensure limited injection to the rest of the Transmission System.

Stakeholders, including those who have developed or plan to develop large loads for interconnection to the MISO Transmission System, also opined that limiting ZG Interconnection Service to a single Point of Interconnection may unduly restrict the utility of the product or unnecessarily increase its cost because Interconnection Customers and Transmission Owners would be denied the flexibility to utilize adjacent bays at the substation for the generation and load lead lines when this is more practical or cost-effective than expanding the single bay that constitutes the Point of Interconnection to accommodate both the load and generation lead lines. The proposed language affords a modicum of flexibility while still maintaining clear and narrow limits.

After considering stakeholder feedback and precedent on generator replacement, MISO determined that allowing the load to connect at the same voltage level of the same substation as the generation would be electrically indistinguishable from requiring them to connect behind the same Point of Interconnection and that with the appropriate equipment, injection to the rest of the Transmission System could be effectively limited in both

scenarios. Based on these considerations and stakeholder opposition to allowing ZG Interconnection Service to address broader proximate load and generation configurations, MISO determined that the proposed scope struck the best balance for the MISO footprint between enabling large loads and protecting the Transmission System from broader impacts.

MISO will continue to work with stakeholders on developing rules tailored to the more dispersed proximate load and generation scenarios contemplated by the Show Cause Order. However, these solutions will be addressed in a separate Federal Power Act Section 205 Filing. The current filing does not impede that separate effort.

Q. WHY DOES MISO PROPOSE ZG INTERCONNECTION SERVICE AS A PERMANENT PRODUCT?

A. MISO proposes ZG Interconnection Service as a permanent product because the service is designed for co-location arrangements in which generation is electrically tied to a specific load at the same substation and voltage level and can be prevented from injecting into the broader Transmission System. A temporary service would create unnecessary costs and uncertainty for generation and load in such configurations. There are several reasons for this determination. First, allowing a permanent service will provide greater up-front financial and operational certainty to the Interconnection Customer seeking ZG Interconnection Service. Second, as I testified earlier, ERIS is a broader product than is needed to enable these types of arrangements. If MISO were to make ZG Interconnection Service temporary, this would result in the ZG Interconnection Customer that intends to only continue to serve its co-located Load, later needing to pay unknown costs to go back through the queue to obtain a service that it does not need and will not use. Third, the

limitations on injections to the rest of the Transmission System mitigate concerns about impacts over time from the ZG Resource with ZG Interconnection Service because the limitations on its injection are a core component of the product. Should an Interconnection Customer seek to change its configuration to allow broader injections, this would require a new Interconnection Request for an appropriate service, such as ERIS or NRIS. Fourth, it would not be efficient or fair to weigh down the DPP with Interconnection Requests that restudy generation and load that do not need to change their previously-studied configuration because a time limit was placed on the original service type. MISO determined that the better limitation is the prohibition on injection. If an Interconnection Customer with ZG Interconnection Service seeks to serve different load or remove the limitations included in ZG Interconnection Service, it may seek ERIS or NRIS service through an appropriate Interconnection Request. Barring such a change, the value of having a durable service-when coupled with the limited configuration options that ZG Interconnection Service authorizes—provides more value to the MISO footprint than scoping ZG Interconnection Service as a temporary but electrically broader product. The appropriate limiting feature for ZG Interconnection Service is the prohibition on injection to the Transmission System, not an arbitrary time expiration. Given that ZG Interconnection Service is continuously bounded by a narrow set of configurations and operational options and is studied to ensure it does not affect the broader Transmission System, a permanent service strikes the appropriate balance between reliability and administrative efficiency.

To be clear, I am not stating that a temporary service is inappropriate for more dispersed resource and load pairs—such as those that resemble the "proximate"

configurations that the Commission describes in the Show Cause Order. In fact, MISO currently is exploring processes that accommodate such broader configurations with stakeholders.

Q. CAN INTERCONNECTION CUSTOMERS COMBINE ZG INTERCONNECTION SERVICE WITH OTHER INTERCONNECTION SERVICE TYPES?

A. No. ZG Interconnection Service is fundamentally different from ERIS or NRIS service. ZG Interconnection Service enables interconnection but excludes injections beyond the substation where the ZG Resource and ZG Associated Load are interconnected. (except for certain inevitable transient injection scenarios that I discuss below), while ERIS allows injections of a specified amount as available, and NRIS allows unrestricted injections of a specified amount. ZG Interconnection Service and the ZGIA require configurations and/or control technology to restrict injections onto the broader Transmission System. Fundamentally, a resource cannot be interconnected with a prohibition on injection and also have the right to inject onto the Transmission System. For this reason, no amount of ERIS or NRIS can be held by a Resource at the same time that it has ZG Interconnection Service.

Q. CAN AN INTERCONNECTION CUSTOMER WITH ZG INTERCONNECTION SERVICE OBTAIN ERIS OR NRIS?

A. Yes. An Interconnection Customer with ZG Interconnection Service is free to submit an Interconnection Request into MISO's queue for ERIS and/or NRIS. Once the Interconnection Customer completes studies and obtains ERIS or NRIS for its ZG Resource, this automatically would supersede any ZG Interconnection Service that such ZG Resource has and the parties would replace or amend the ZGIA with a GIA reflecting

the new service levels. This process applies, even where the Interconnection Customer obtains a lesser megawatt amount of ERIS or NRIS than the ZG Interconnection Service it previously had. The reason for this is because the prohibition on injections and requirement for equipment that electrically isolates the ZG Resource from the broader Transmission System is being replaced with an ability to inject at some level. In such a circumstance, the ZG Resource would then be treated as any other Generating Facility that has obtained Interconnection Service below its installed capacity.

Q. PLEASE PROVIDE AN EXAMPLE OF THIS SCENARIO.

A. Assume a Generating Facility obtains ZG Interconnection Service for its full 500 MW of installed capacity in order to serve a ZG Associated Load that is behind the Point of Interconnection of the Generating Facility. The Generating Facility would receive a 500 MW ZGIA and its appendices would describe the equipment necessary to limit injection to the rest of the Transmission System. Several years later, the Interconnection Customer of this service decides that it wants to use same Generating Facility to serve loads elsewhere on the Transmission System at certain times when the Demand associated with its co-located load may be reduced by 300 MW. The Interconnection Customer would be able to submit an Interconnection Request for 300 MW of ERIS or NRIS and proceed through MISO's queue (i.e., DPP) as would any other Interconnection Request. At the conclusion of the DPP, the Interconnection Customer would obtain a 300 MW GIA for ERIS or NRIS. The appendices to such GIA would describe the Generating Facility, including its installed capacity, as well as any metering or other equipment required under MISO's ordinary Tariff processes to facilitate such service below capacity. This would replace the requirement to maintain equipment that limits all injections beyond the substation

described in the ZGIA and the corresponding descriptions of such equipment in the appendices. Under such a configuration, the Generating Facility may still be allowed to provide energy to the co-located load above the 300 MW Interconnection Service level because such energy would not enter the Transmission System and the GIA would appropriately bind such Interconnection Customer to MISO's Tariff rules, including rules governing unauthorized use of the Transmission System.

Q. DOES THE PROCUREMENT OF TRANSMISSION SERVICE BY ZG ASSOCIATED LOAD DIFFER FROM THE PROCUREMENT OF TRANSMISSION SERVICE BY LOAD SERVED BY GENERATORS THAT ENTER THE DPP?

A. No. The ZGIA facilitates the interconnection of a generator at the same substation as ZG Associated Load. Like all MISO interconnection agreements, a ZGIA provides the interconnection service to a resource, but does not provide Transmission Service. Using language already in MISO's *pro forma* GIA, the ZGIA explicitly states that it does not grant or otherwise convey Transmission Service to the ZG Associated Load. The current filing does not establish different rules regarding obtaining transmission service from those that generally apply to load. I am aware that the Show Cause Order contemplated changes to transmission service for co-located, proximately located, and other large load scenarios. MISO intends to address those matters in a separate filing.

Q. PLEASE DESCRIBE HOW THE ZGIA AND ZGIA PROCESS FIT INTO MISO'S BROADER LARGER LOAD ENABLEMENT EFFORT.

A. The ZGIA construct is one tool to enable speed to power for generation, specifically generation that is located geographically and electrically close to the load it plans to serve.

A ZGIA enables these uniquely situated generators to avoid the costs and time of unnecessary participation in the queue, provided impacts to the rest of system can be isolated. Further, the framework provides visibility into generation behavior in circumstances where MISO may not otherwise have that visibility due to lack of a GIA. Speed to power is achieved because the framework allows for the ZG Associated Load – which in many cases may be a large load - to have generation modelled when being studied in the MISO Transmission Expansion Plan ("MTEP"). The ZGIA construct is also a step towards integrated transmission and generation studies through tighter coordination between the generation and load.

Q. PLEASE ELABORATE ON HOW THE ZGIA PROCESS PROVIDES SPEED TO POWER TO LARGE LOAD ENABLEMENT.

A. Only resources that have an accepted GIA are eligible for inclusion in MTEP studies. MISO's MTEP process–including its Expedited Project Review ("EPR") process, that provides a clear picture of network upgrade facilities and costs required to integrate a new load–can move on a 90-day timeline. However, the MISO interconnection queue does not move this fast. The ZGIA process will provide an interconnection customer with an agreement within 90 days. With an interconnection agreement in hand, the ZGIA Resource can then be modeled as an input to the subsequent MTEP (or MTEP EPR) load study for its ZG Associated Load. The ZGIA Process provides a much faster timeline for studying large load integrations and creates the potential for reduced total cost of transmission network upgrades needed to integrate the new load, given the recognition of additional nearby generation in the MTEP (or MTEP EPR) study.

Q. DESCRIBE THE CHANGES MADE TO ATTACHMENT FF OF THE TARIFF TO SPECIFY THE RELATIONSHIP BETWEEN A ZG RESOURCE AND ITS ZG ASSOCIATED LOAD.

A. Attachment FF provides for, among other things, the process of studying the Network Upgrades needed to integrate new loads onto the Transmission System. This process includes the capability for Non-Transmission Alternatives, such as Resources (and in this case, a ZG Resource) to offset the need for Network Upgrades. In order to ensure that the load integration study is an accurate representation of future conditions, there must be certainty that the Non-Transmission Alternative becomes operational and remains operational. For this reason, language has been added to Attachment FF that requires an executed agreement between the ZG Resource and the ZG Associated Load, that the term of the agreement be at least five years which aligns with the MTEP planning horizon, and that allows the Transmission Provider to require the delay of the in-service date of the ZG Associated Load until alternate Network Upgrade solutions, needed due to the failure of the ZG Resource as a Non-Transmission Alternative, can be put in place.

Q. WHAT ARE THE DIFFERENCES BETWEEN MISO'S APPROACH AND SPP'S HILLGA APPROACH?

A. Fundamentally, HILLGA and ZG Interconnection Service serve the same goal: allowing for the rapid study and development of generation needed on an accelerated basis to serve a specific load need. Both constructs balance a need for rapid study with protection of the Transmission System and avoidance of negative impacts to other Interconnection Customers. Both also rely on proximity of the generation to the load to minimize impacts to the broader system. In minimizing the potential for unexpected impacts to the

Transmission System, there are two design considerations that can be scoped in relation to each other to achieve balance: (1) duration of service, and (2) narrowness of design scope. The more of the transmission system that is allowed to be used (i.e., the permitted electrical remoteness of the generation and load), the greater the potential for impacts to the system to change in unexpected ways. This is because it becomes harder to truly isolate injections. The main protections against system impacts thus are reliable modelling and the consistency of the generation and load to the modelling assumptions. This is more of a short-term focused solution. The longer that service is allowed to remain, the more it provides upfront cost certainty to Interconnection Customers. But the tradeoff is that the system will change over time and impact assumptions may change with it. Shorter term service like SPP's HILLGA allows for impacts to be mitigated or made temporary because the service will be replaced with something permanent after only a few years. A permanent product, like ZG Interconnection Service relies on tight geographical requirements (i.e., co-location of the ZG Resource and ZG Associated Load at the same substation) and isolation from the rest of the Transmission System (i.e. prevention of injection) to ensure that even if the Transmission System changes over the course of decades, the impact of the ZG Resource will remain similar to the initial impacts modeled as part of the ZGIA study process.

Q. WHY IS MISO PROPOSING THE ZGIA CONSTRUCT AS A PERMANENT PRODUCT INSTEAD OF A SHORTER-TERM PRODUCT?

A. MISO selected the ZGIA approach to promote an efficient interconnection process and accelerate service to load, while also preserving equitable treatment of interconnection customers and accommodating preexisting arrangements. Stated differently, the ZGIA

framework balances the requirement that generators injecting onto the Transmission System be studied and appropriately mitigated with the recognition that certain on-site generation serves local load without exporting power to the broader Transmission System. The ZGIA approach builds from experience in the MISO footprint and recognizes both stakeholder preferences and the types of co-location and near co-location configurations that MISO has encountered and expects to see in the future.

Q. WHY IS IT IMPORTANT TO HAVE A GIA FOR A GENERATOR THAT DOES NOT PLAN TO INJECT ENERGY ONTO THE TRANSMISSION SYSTEM?

A. Historically, MISO has only rarely encountered Interconnection Customers seeking to connect to the Transmission System without injecting energy. However, with the advent of significant large load additions, MISO has seen these types of arrangements become more common and expects this trend to continue. Therefore, it is now important to implement a service type and study process that specifically recognizes both the challenges and opportunities associated with large loads, leverages the increased commercial attractiveness of pairing generation with a specific nearby load, and tailors an Interconnection Service study process to their needs. The proposed ZG Interconnection Service and ZGIA framework provide a critical first step to better enabling large loads while also maintaining equitable interconnection principles–resource injections on the grid must be studied, and their impacts to prior-queued and operational resources must be mitigated, and other Interconnection Requests should not be harmed.

If a Generating Facility is connected to MISO's Transmission System and has the ability to inject energy onto the Transmission System, MISO must have visibility into how that Generating Facility is configured and will operate in order to determine whether the

interconnection poses reliability risks to the Transmission System, regardless of whether the Generating Facility intends to inject energy onto the Transmission System. Relatedly, while the owner of a Generating Facility may state an intention not to inject onto the MISO Transmission System, an agreement is required to ensure that such an intention becomes a binding commitment to promote the reliable operation of the Generating Facility and provide recourse if the Generating Facility operates contrary to its stated intent.

As a practical example, MISO's Resource Utilization team has encountered several situations where the owner of a generating facility claimed that its equipment or configuration actually affirmatively prevented injections, thus removing the need for Interconnection Service. However, absent a generator interconnection agreement or similar form of agreement, MISO would have no way to require the Generating Facility to be constructed as represented to MISO. This includes ensuring that sufficient protection equipment is in place to prevent injection onto the Transmission System, as contemplated by the Show Cause Order. Given the risks to Transmission System reliability of addressing unauthorized injections only after they happen or attempting to impose operational limitations on a Generating Facility once it is already constructed, MISO determined that it is both prudent and necessary to require the developers of Generating Facilities that connect to the MISO Transmission System to become Interconnection Customers and execute a GIA to establish necessary safeguards upfront. However, MISO recognizes that these Generating Facilities were not intended to inject energy onto the Transmission System. It therefore was neither necessary for them to request an amount of ERIS or NRIS Interconnection Service above 0 MW nor fair to make them enter MISO's queue simply to

validate and memorialize a commitment not to inject. The ZGIA provides a targeted solution to this problem.

IV. THE PROPOSED ZGIA CONSTRUCT

Q. PLEASE DESCRIBE ZG INTERCONNECTION SERVICE.

A. ZG Interconnection Service is service limited to serving specifically designated load that is either at the same Point of Interconnection ("POI") or has a Point of Withdrawal ("POW") on the same voltage side of the same substation as the ZGIA. Like other forms of Interconnection Service in MISO, ZG Interconnection Service is not Transmission Service and does not exempt a transmission customer, including a transmission customer serving a large load, from procuring Transmission Service.

Q. WHAT IS A POINT OF WITHDRAWAL AND HOW DOES IT RELATE TO THE ZGIA?

A. A Point of Withdrawal is a newly-defined term that describes the point that a Load interconnects to the Transmission System. It is the Load analog to the generation-specific term "Point of Interconnection". It is important that this connection point is specified with respect to ZG Interconnection Service, because resources are limited to taking ZG Interconnection Service only at the same substation and voltage level of the Point(s) of Withdrawal of their ZG Associated Load.

Q. PLEASE EXPAND ON HOW ZG INTERCONNECTION SERVICE IS A LIMITED PRODUCT.

A. As noted below, ZG Interconnection Service allows the Interconnection Customer to connect at the substation and voltage level of its ZG Associated Load. ZG Interconnection Service does not allow the Interconnection Customer to serve other load or inject energy

onto the Transmission System for purposes other than serving the co-located ZG Associated Load identified in the Interconnection Request and memorialized in the ZGIA.

The study requirements necessary for a ZGIA to provide ZG Interconnection Service are robust, including obligations to implement equipment and/or operational procedures to prevent injections onto the Transmission System beyond the substation where the ZG Resource and ZG Associated Load are located, and to limit any inadvertent injections to a defined Transient Tolerance Window, as described below, in the event of an unexpected load reduction or trip.

Q. WILL ZG INTERCONNECTION SERVICE BE AVAILABLE ONLY TO GENERATORS SERVING LARGE LOADS?

A. No. While the tool is particularly responsive to the needs of large loads, a ZG Associated Load does not have to be a large load.

Q. IS THERE A LIMIT ON THE NUMBER OF ZGIA PROJECTS STUDIED PER CYCLE?

A. Yes. MISO will process the ZGIA Interconnection Requests outside of its Definitive Planning Phase using a serial "first-come, first-served" approach based on the date and time the application was submitted. The first ten projects to submit applications to be studied in the ZGIA process will be tentatively selected until MISO has finalized its screening process of all ten projects to ensure that they meet the ZGIA eligibility requirements. After the ten projects in a cycle are finalized, the remaining projects that applied in the cycle will be deferred to the next available ZGIA cycle in order of submission time. This limit was developed to prevent disruption to MISO's DPP, and to ensure that staffing resources are available.

Q. PLEASE DESCRIBE HOW MISO WILL MONITOR AND LIMIT INJECTIONS.

A. Among other elements, The ZG Interconnection Study includes a review of the control equipment proposed by the Interconnection Customer to determine whether the proposed configuration, equipment, and/or operating procedures are sufficient to prevent injections onto the Transmission System beyond the voltage level and substation where the Point of Interconnection is located, within a Transient Tolerance Window.

Q. PLEASE DESCRIBE WHY MISO IS UTILIZING A TRANSIENT TOLERANCE WINDOW?

A. The Transient Tolerance Window is a requirement for the ZG Resource to disconnect or reduce output if the ZG Associated Load trips or has a significant load reduction. In this event, if the same breaker does not supply both the ZG Resource and the ZG Associated Load, the Resource will not automatically disconnect along with the ZG Associated Load. Therefore it is not realistic to prohibit all transient injections. Some time period must be allowed for the generator to ramp down or disconnect to match the demand of the ZG Associated load. During this time, incidental flows onto the Transmission System can occur. The Transient Tolerance Window provides a required time frame for the ZG Resource to disconnect or match ZG Associated Load demand.

Q. WHY DID MISO CHOOSE 20 SECONDS AS THE TRANSIENT TOLERANCE WINDOW?

A. The Transient Tolerance Window is limited to values less than or equal to 20 seconds in order to align with the MTEP modeling interval of 20 seconds for transient stability analysis. This ensures that a generator will be robustly evaluated for stability impacts to ensure that it will operate without unnecessary tripping for momentary conditions that do

not affect the Transmission System. Transmission Owner Local Planning Criteria could further limit the Transient Tolerance Window to values less than 20 seconds.

Q. PLEASE DESCRIBE THE ELIGIBILITY REQUIREMENTS FOR ZG INTERCONNECTION SERVICE.

A. MISO has developed a balanced set of requirements for participation in ZGIA to ensure that the accelerated study is utilized for projects that are commercially viable. These requirements were significantly shaped by stakeholder discussions and feedback. As such, MISO is proposing the following technical requirement for ZGIA that must be met at the time of application submission: The Interconnection Request must include a model and one-line in which both the ZG Resource and load must be independently represented. This enables MISO to study the generation in connection with the ZG Associated Load and verify the sufficiency of the configuration and proposed control equipment to limit injections.

- The Interconnection Request must specify a Point of Interconnection and Point of Withdrawal at the same voltage level of the same substation.
- A $150,000 non-refundable D1 fee must be paid by the Interconnection Customer to facilitate processing.
- The amount of ZG Interconnection Service applied for is limited to and shall not exceed the total Demand of the ZG Associated Load for the proposed ZG Resource.
- 100% Site Control for the ZG Resource as well as 100% Interconnection Customer's Interconnection Facilities ("ICIF") Site Control will be required when the application is submitted.

- The Interconnection Request for ZG Interconnection Service shall be accompanied by an executed agreement evidencing that the full output of the ZG Resource for which ZG Interconnection Service is sought is intended to be used by the entity(s) with the ZG Associated Load with a minimum term of five years in order to allow it to be relied upon in MTEP.
- The submitted Interconnection Request cannot include a request for any other type of Interconnection Service (e.g., ERIS or NRIS).
- The ZG Interconnection Studies cannot identify the need for any Network Upgrades beyond the substation where the Point of Interconnection is to be located other than breakers on electrically adjacent substations.

Q. HOW IS ZG ASSOCIATED LOAD IDENTIFIED AND STUDIED IN THE ZG INTERCONNECTION PROCESS?

A. Section 3.11 of the proposed GIP requires an Interconnection Request for ZG Interconnection Service to include a one-line diagram that includes the Generating Facility and associated electrical equipment with appropriate rating and impedance information and all the ZG Associated Load including all Point(s) of Withdrawal and the Point of Interconnection of the Generating Facility to the Transmission System for the Interconnection Request. It also requires a detailed power flow model for the proposed Generating Facility that includes all ZG Associated Load, Point(s) of Withdrawal and the Point of Interconnections for the Interconnection Request, as well as any proposed system protection facilities. Inclusion of this detailed modeling of the ZG Associated Load will ensure that the interconnection study has flows and impacts that are relevant given the known relationship and co-location of the ZG Resource and its ZG Associated Load.

The ZG Interconnection System Impact Study will generally follow the already established studies and criteria for DPP studies (steady state, stability and short circuit). However, one important difference will be that during the ZG Interconnection System Impact Study, energy from the Generating Facility is modeled as serving the ZG Associated Load rather than injecting into the Transmission System as under a System Impact Study that assumes ERIS. However, the study will also include a scenario where the load is tripped in the stability analysis. This is to ensure that the Transmission System remains reliable under the scenarios where the ZG Resource is required to curtail when the ZG Associated Load is tripped.

Q. WHY IS THE ZG INTERCONNECTION CUSTOMER REQUIRED TO MOVE ITS POINT OF INTERCONNECTION IF THE ZG ASSOCIATED LOAD MOVES?

A. This requirement is essential to ZG Interconnection Service due to the required electrical relationship between the ZG Resource and the ZG Associated Load, and the serial nature of the relevant integration studies. Because the ZG Interconnection Study must take place before the load integration study of the ZG Associated Load, it is possible that the ZG Associated Load may decide or be required to move its Point(s) of Withdrawal in some way after ZG Interconnection Studies are in process or completed. Ordinarily, an Interconnection Customer attempting to move its Point of Interconnection after studies have begun would be subject to material modification rules due to the fact that changing the Point of Interconnection alters electrical flows on the Transmission System. With ZG Interconnection Service, the mechanism for avoiding impacts to the rest of the Transmission System is requiring the Generating Facility and the ZG Associated Load to

connect at the same voltage level and substation with injections to the rest of the Transmission System restricted. Because it is not practical to prevent the ZG Associated Load from moving before its studies have even begun, it is necessary to ensure that the Generating Facility follows any movement of the Load and remains at the same substation as the ZG Associated Load.

Q. WILL ACTIVE DPP PROJECTS BE ABLE TO TRANSFER TO A ZGIA?

A. Yes. If the project meets the eligibility requirements to enter the ZGIA interconnection process, DPP projects in active DPP cycles at or before Decision Point 2 may transfer to the ZGIA process. MISO will not allow Interconnection Requests to transfer from the existing interconnection queue after Decision Point 2 to avoid Phase restudies due to late withdrawals, which contribute to ongoing queue delays. If an Interconnection Request is beyond Decision Point 2, then it can still drop out of the DPP process; however, it cannot submit a ZGIA until a year has passed since the withdrawal. All projects withdrawing from the DPP process and entering ZGIA will be subject to all existing DPP penalties and harm calculations as outlined in the Tariff. This approach balances flexibility for projects in earlier stages of the DPP to instead seek ZG Interconnection Service as the service better suited to their needs while discouraging late dropouts after established DPP Decision Points, which can disrupt the timing and expectations of other Interconnection Customers.

Q. PLEASE DESCRIBE HOW INTERCONNECTION SERVICE GRANTED THROUGH A ZGIA WILL BE STUDIED.

A. As explained previously in my testimony, Interconnection Requests seeking ZG Interconnection Service will be studied in a serial “first-come, first served” fashion based on the order the projects were submitted. It will follow a quarterly cadence established by

proposed MISO timelines, ultimately culminating in the project receiving a ZGIA in approximately 90 days. During the ZGIA model build, all Existing Generating Facilities, and the latest MTEP base case with all approved MTEP projects, including any recently approved EPR projects will be utilized as the starting point for the ZG Impact Analysis. This is the same process utilized for the DPP, except the DPP models include all higher queued Interconnection Requests from early cycle and all Interconnection Requests within that DPP cycle. ZGIA studies will not include Interconnection Requests currently in the generator interconnection queue that do not have a GIA.

Before every ZGIA cycle, all Interconnection Requests with a GIA approved since the last base case was created will be added to the base case for the following ZGIA cycle but are not included in any ongoing DPP studies. The base case for each DPP cycle is never updated once the study process is kicked off except for generation changes due to Interconnection Request withdraws or generator retirements. This ensures that the transmission capacity made available for DPP projects is not being taken away by ZGIA Interconnection Requests approved outside of the DPP cycle.

Once the DPP cycle has concluded, the output from that process and any completed ZGIA and EPR projects will be used to populate a new MTEP base case. Any transmission issues identified by reconciling the cases will thus be addressed through the existing MTEP process. The impacts from the ZGIA are expected to be minimal given the requirements for ZG Interconnection Service. The next DPP cycle will then use the latest approved MTEP base case that has reconciled the models. This insulates existing DPP projects from negatives impacts or cost shifts due to the existence of the ZGIA process. The contingent

facilities listed in the ZGIA would only be those facilities in the model used for the ZGIA study.

The ZGIA study process will resemble the DPP study process and criteria with two important changes. First, ZG Interconnection Service will not be granted if a Network Upgrade is required beyond the substation where the Point of Interconnection is located other than System Protection Facilities. Second, the study will include a simulation of the scenario where the ZG Associated Load trips offline. This is to ensure that specific protection schemes are validated for the ZGIA project. These two changes are needed to reflect the limited nature of the ZG Interconnection Service.

Q. PLEASE EXPLAIN WHY ONLY CERTAIN NETWORK UPGRADES WILL BE ALLOWED WITH THE ZG STUDY?

A. Working together with stakeholders, MISO developed Network Upgrade guardrails that ultimately prevent injections on the Transmission System beyond the substation where the Point of Interconnection is located, and ensure that other Interconnection Customers are not impacted by the interconnection of a ZG Resource. Interconnection of a ZG Resource must not require the construction of Network Upgrades except for Network Upgrades at the substation of the POI, and System Protection Facilities that could be required beyond the substation where the Point of Interconnection is located. This limits the Network Upgrades (other than System Protection Facilities) to only those introduced as part of the ZGIA project itself—that is, facilities needed to connect the ZG Resource to the Transmission System. For example, new equipment at the bay of the Point of Interconnection may be required to accommodate the Resource. The reason that system protection facilities on neighboring substations are allowed is because the reliable

operation of the Transmission System may require installation of a breaker or similar equipment and, from a practical engineering perspective, the proper place for such equipment may be at the neighboring substation rather than the substation where the POI is located. Installation of such Network Upgrades is consistent with the goal of ZG Interconnection Service of limiting injections and impacts to the rest of the Transmission System. Because such equipment may be the best method of limiting injections and is limited in scope, we determined that otherwise viable Interconnection Requests seeking ZG Interconnection Service should not be disqualified on the basis of such limited Network Upgrades.

Q. WILL ZG RESOURCES NEGATIVELY IMPACT EXISTING PROJECTS IN THE GENERATOR INTERCONNECTION QUEUE?

A. No. Under this model framework, the standing MISO DPP three-phase process will not be impacted by the ZG Interconnection Service Study Process. This is true for several reasons. First, because a ZG Resources cannot inject beyond the substation where it has a POI and because it cannot be approved if it triggers the need for Network Upgrades on other parts of the Transmission System, effectively limiting their potential interaction with other Interconnection Requests. Second, Interconnection Requests seeking ZG Interconnection Service will not be included in ongoing DPP studies, thereby not utilizing any available transmission capacity in those studies. The normal DPP process uses the most up to date MTEP base case when the study is kicked off and is not modified to include later approved MTEP projects or ZGIAs. The models from ZG Interconnection Studies and the DPP will be reconciled through MTEP.

Q. PLEASE PROVIDE SOME EXAMPLES OF OPERATING PROCEDURES CONSISTENT WITH PROPOSED SECTION 3.2.5.2.

A. MISO anticipates reviewing a variety of configurations, physical equipment, and operational requirements that may independently, or in sum, prevent injections on the Transmission System beyond the substation where the Point of Interconnection is located. For example, the injection limitation may be implemented through equipment controls such as export-limiting controls, directional relays, interlocks, and fail-safe trip logic. Operational requirements may include generation limits, curtailment obligations, outage coordination, and restrictions on operation. These operational requirements are in addition to any monitoring, testing, or maintenance requirements. It is important to note that Transmission Owner Local Planning Criteria could serve to further limit the set of feasible approaches that the Interconnection Customer can propose to limit injections.

Q. HOW WILL THE 10 PROJECTS BE SELECTED FOR THE STUDY?

A. Once the ZGIA process is approved, MISO will process the ZGIA Interconnection Requests using a serial "first-come, first-served" approach based on the date and time the application was submitted. The first ten projects to submit applications to be studied in ZGIA will be tentatively selected until MISO has finalized its screening process of all ten projects to ensure that they meet the ZGIA eligibility requirements. After the ten projects in a cycle are finalized, the remaining projects that applied in the cycle will be deferred to the next available ZGIA cycle in order of submission time. Since only ten projects will be studied at a time, serial review of the applications will expedite the review process. MISO expects the ZGIA Interconnection Requests to apply when they are ready and meet all the

necessary eligibility requirements to participate. All application deadlines and ZGIA Interconnection Requests will be posted on the MISO public website in a timely fashion.

Q. WHY DID MISO CHOOSE TO USE A OFFTAKE AGREEMENT PROCESS SIMILAR TO THAT USED IN THE ERAS PROCESS?

A. Because the ZG Resource can only serve its ZG Associated Load, the ZGIA process requires a clear contractual relationship between the two. The set of acceptable contractual arrangements that is currently effective for the ERAS process serves that role well, with one exception. Since some jurisdictions allow resource ownership directly by the retail customer, an acceptable contractual arrangement could be "an acknowledgement by the entity that owns the ZG Associated Load that such entity is both permitted by applicable law and intends to self-supply its ZG Associated Load through using onsite generation".

Q. WILL ZG INTERCONNECTION REQUESTS GO THROUGH THE AFFECTED SYSTEM EVALUATION PROCESS?

A. Yes. While MISO does not anticipate a ZG Resource to cause impacts to Affected Systems, due to the nature of the service, it is reasonable for Affected Systems to have the opportunity to study ZG Resources for impacts if they believe it necessary. The language proposed in Section 3.5.3 of the GIP establishes a framework for how such studies are coordinated in a similar fashion as other Interconnection Requests. The affected system likely would not need to engage in more than a limited review to verify that injections are limited and either do not reach the affected system or are addressed through system protection facilities at the neighboring substation. The language allows flexibility for MISO to work with its seams partners to develop any ZG Interconnection-specific process

refinements that are mutually agreeable while also providing a reasonable default framework.

Q. HOW DID MISO ACCOUNT FOR OPEN ACCESS REQUIREMENTS WHEN DEVELOPING THE ZG PROCESS?

A. The ZGIA requirements are facially neutral, fuel-agnostic, and allow for the potential inclusion of any resource regardless of technology. Any Interconnection customer with ZG Associated Load may be studied for ZG Interconnection Service regardless of their identity as long as they meet the applicable state's legal requirements.

Q. CAN A RESOURCE WITH ZG INTERCONNECTION SERVICE PARTICIPATE IN THE REPLACEMENT PROCESS?

A. Yes. Once a ZG project receives a ZGIA, it is eligible to participate in the replacement process under the same rules as other Interconnection Requests. However, because these facilities would have ZG Interconnection Service, they could only replace it with another Generating Facility that has ZG Interconnection Service in the same location. Should such Interconnection Customer seek a different service type, they would need to submit an Interconnection Request through the DPP.

Q. WILL ZGIAS BE SUBJECT TO MISO'S REGIONAL PLANNING PROCESS?

A. Yes. Resources seeking ZG Interconnection Service will be subject to the specific local planning criteria of the MISO Transmission Owners. These are based on applicable reliability standards such as NERC FAC-002 and outline the requirements that each Transmission Owner deems necessary to maintain local reliability within their planning processes.

V. ZG RESOURCE PARTICIPATION IN RESOURCE ADEQUACY

Q. HOW DOES THE ZGIA CONSTRUCT SUPPORT RESOURCE ADEQUACY IN THE MISO REGION?

A. As detailed in MISO's Information Report on Resource Adequacy submitted on June 18, 2026, in this docket, MISO continues to work diligently with the states both individually and through the Organization of MISO States ("OMS"), as well as its member utilities, to support resource adequacy for planned load growth in a manner that maintains customer affordability, efficiency and regional reliability. The ZGIA construct is particularly responsive to the challenges presented by large loads additions as it will facilitate the accelerated interconnection of generation needed to serve a specific nearby load source where broader injection onto the broader Transmission System is not needed and system impacts—and corresponding Network Upgrades—can be limited to roughly the substation level. This provides a valuable option for generation to be co-located with large loads without having to enter the DPP of MISO's generation interconnection queue or obtain a form of Interconnection Service that is broader and potentially more costly than what is needed.

Q. CAN ZGIAS PARTICIPATE IN MISO'S RESOURCE ADEQUACY CONSTRUCT?

A. Yes. A ZGIA Resource can participate as a Generation Resource, Dispatchable Intermittent Resource, or Electric Storage Resource, and in some cases, as Behind the Meter Generation; thereby qualifying them as a Capacity Resource. Participation eligibility and participation obligations are subject to the currently effective Tariff Module E-1.

Q. HOW WILL A ZG RESOURCE'S ACCREDITATION VALUE BE DETERMINED?

A. ZG Resources will be accredited like any other Capacity Resource of the appropriate technology type except that a ZG Resource will offset no more than the coincident peak demand of the ZG Associated Load, plus an accounting for system losses not experienced due to the co-location of load and generation. ZG Resources will not be able to offset any other loads, because the ZG Resource is not able to inject onto the Transmission System.

Q. IS A ZG RESOURCE THAT PARTICIPATES IN MISO'S RESOURCE ADEQUACY CONSTRUCT SUBJECT TO A MUST OFFER REQUIREMENT?

A. Yes. As a Capacity Resource, a ZG Resource will be subject to the "must offer" obligation into the MISO's Day-Ahead Energy Markets when it clears the PRA, with the exception that the must-offer obligation is limited to the expected demand of the ZG Associated Load for the next Operating Day.

Q. HOW IS A ZG RESOURCE ABLE TO PARTICIPATE IN MISO'S RESOURCE ADEQUACY CONSTRUCT?

A. To ensure that the limitations of the ZG Resource are respected, and to prevent the potential for ZRC sales from a ZG Resource to a load other than the ZG Associated Load, ZGIA Resources must be in a Fixed Resource Adequacy Plan ("FRAP") or RBDC Opt-Out of the Market Participant that has the Planning Reserve Margin Requirement of the ZG Associated Load. Further, ZG Resource are not eligible to replace ZRCs throughout the Planning Year.

VI. ZG RESOURCE PARTICIPATION IN THE ENERGY AND OPERATING RESERVE MARKETS

Q. ARE ZGIA RESOURCES REQUIRED TO REGISTER AND PARTICIPATE IN MISO MARKETS?

A. ZGIA Resources that are greater in size than 5 MW will be required to register and participate as either Generation Resources, Dispatchable Intermittent Resources, or Energy Storage Resources in the MISO Energy and Ancillary Service Markets. This requirement ensures that ZG Resources and their ZG Associated Loads are allocated a share of market-distributed costs in proportion to the benefits received. Certain market charges such as the provision of ancillary services for load, and administrative charges, are billed based on gross MW volumes in the Day-Ahead and Real-Time markets. By requiring ZGIA Resources to register as market resources, MW volumes for both load and generation will be captured separately, resulting in appropriate billing of these charges under already-applicable Tariff rules for market settlements.

Q. WHY ARE ALL ZGIA RESOURCES NOT REQUIRED TO REGISTER AND PARTICIPATE IN MISO MARKETS?

A. MISO proposes not to require any ZG Resource smaller than 5 MW in size to register and participate as a market resource, because this is the de minimis size threshold that MISO typically provides for Resource registration.

Q. HOW WILL A ZG RESOURCE MAINTAIN ITS OUTPUT BELOW THE CONSUMPTION OF THE ZG ASSOCIATED LOAD WHILE RESPONDING TO DISPATCH INSTRUCTIONS FROM MISO?

A. MISO's Day-Ahead and Real-Time markets create economically efficient dispatches to generators that account for the physical and economic capabilities of resources. ZG Resources are required to submit maximum limit offers that are at most equal to the demand

expectation of the ZG Associated Load. With this requirement, the Offers of a ZG Resource will have to be updated throughout the Operating Day. With this maximum limit in place, the dispatch instruction from MISO will be less than or equal to the offsetting demand of the ZG Associated Load, with no resulting injections.

Q. CAN A ZG RESOURCE QUALIFY TO PROVIDE OPERATING RESERVES?

A. Yes. A ZGIA Resource that meets the eligibility requirements can provide operating reserves. Operating Reserves are cleared when energy dispatches of a resource are between its maximum and minimum limits, and Operating Reserve deployments do not result in resource setpoint instructions exceeding the maximum limit. By requiring a maximum limit that is less than the demand expectation of the ZG Associated Load, operating reserve deployments will not result in injections**.** Again, the Offers of a ZGIA Resource will have to be updated throughout the Operating Day to reflect the expected demand of the ZG Associated Load

Q. WILL ZG RESOURCES BE ELIGIBLE TO PROVIDE REACTIVE POWER?

A. Yes. Reactive power does not perform the useful work of the power grid; instead, it supports the system voltage and magnetic fields necessary for reliable operation of the grid. The need for reactive power varies by location and system conditions. Because the provision of reactive power is distinctive from the injection of real power, ZG Resources will be eligible to provide reactive power.

VII. STAKEHOLDER PROCESS

Q. PLEASE DESCRIBE THE STAKEHOLDER PROCESS USED TO DEVELOP THIS PROPOSAL.

A. MISO has engaged in detailed discussions with stakeholders on Large Load issues since 2024 through various stakeholder entities and workshops. MISO first discussed Large Loads with stakeholders in the context of EPR process improvements at the Planning Advisory Committee ("PAC") and Planning Subcommittee ("PSC") in 2024 and 2025. Moreover, since 2024, MISO has discussed coordinating the MISO Transmission Expansion Plan and Definitive Planning Phases processes to accommodate Large Loads at the PSC.

MISO first introduced the ZGIA concept to stakeholders at the PSC on December 3, 2025. At the meeting, MISO provided a high-level overview of the ZGIA concept and solicited feedback from stakeholders. MISO subsequently discussed the proposal at the PAC and the Large Load Working Group. At the January 21, 2026 PAC, MISO described how ZGIA resource configurations may lead to reduced transmission network upgrade needs and detailed the stakeholder feedback submitted in response to the December 3, 2025 PSC presentation. At the February 25, 2026 PAC, MISO began laying out a proposal for formalizing ZGIA by providing the interconnection and planning elements of the proposal, the first two elements of the proposal. MISO requested stakeholder feedback on the proposal. At the first LLWG meeting on April 20, 2026, MISO described the elements of the ZGIA proposal relating to resource adequacy and markets and reliable operations, the last two elements of the proposal, and requested stakeholder feedback. MISO also provided materials outlining MISO's response to previously submitted stakeholder feedback.

At the April 22, 2026 PAC, MISO shared draft Tariff language with stakeholders reflecting the interconnection and planning elements of the ZGIA proposal. The draft

Tariff language included specifics that originated from stakeholder feedback, including language ensuring that a ZGIA disconnects if the facilities connecting it to its associated load disconnects and language requiring ZGIA zonal resource credits to be included in a fixed resource adequacy plan. MISO also provided materials outlining MISO's response to previously submitted stakeholder feedback. MISO requested stakeholder feedback on the proposed draft Tariff language.

At the May 14, 2026 LLWG, MISO shared draft Tariff language reflecting the elements ZGIA proposal relating to resource adequacy and markets and reliable operations. The draft Tariff language incorporated stakeholder feedback regarding planning studies and multiple Points of Injection. Specifically, in response to stakeholder feedback regarding MISO's initial proposal allowing multiple POIs, MISO proposed only allowing multiple POIs if the ZGIA is located at one of the stations, i.e. Points of Withdrawal, of a large load. MISO requested stakeholder feedback on the draft Tariff language.

At the May 27, 2026 PAC, MISO shared updated draft Tariff language with stakeholders on the interconnection and planning elements of the ZGIA proposal. The draft Tariff language incorporated stakeholder feedback, including language specifying further detail of the ZGIA interconnection studies, language clarifying that ZGIAs do not supersede state jurisdiction, and revising the POI rules. MISO additionally decided not to pursue the campus approach in response to stakeholder feedback. MISO requested stakeholder feedback on the proposed draft Tariff language. At the June 10, 2026 LLWG, MISO reviewed the final ZGIA proposal with stakeholders and provided draft Tariff language that incorporated stakeholder feedback. At the June 29, 2026 LLWG, MISO continued the dialogue regarding the ZGIA proposal to provide stakeholders an opportunity to

provide feedback on any remaining concerns and in response to the Show Cause Order. Additional presentations were made at the July 15, 2026 PAC and the August 12, 2026 LLWG and updated Tariff language was presented to stakeholders on the same day. A table listing the stakeholder meetings, including links to the presentations, is provided as Attachment A hereto.

Q. WHAT WAS MISO'S RESPONSE TO CONCERNS AND RECOMMENDATIONS FROM STAKEHOLDERS?

A. As described above, throughout the stakeholder engagement period, MISO has solicited feedback from stakeholders and provided written and verbal responses to that feedback. MISO's stakeholder process has contributed to substantive design changes in MISO's filing and the collaborative process with stakeholders resulted in a comprehensive ZGIA proposal. MISO appreciates the willingness of all its stakeholders and state regulators to review and comment on this final proposal, which MISO believes reasonably accommodates concerns raised by stakeholders. Due to stakeholder feedback, MISO modified the ZGIA proposal by reducing the geographic scope of allowable ZGIA interconnection points, the specific requirements and expectations of the ZG Interconnection study, and accreditation value of ZG Resources.

Q. DID MISO MODIFY ITS ZGIA PROPOSAL DUE TO THE SHOW CAUSE ORDER ISSUED BY THE FEDERAL ENERGY REGULATORY COMMISSION?

A. Yes. Given both stakeholder feedback, and concerns raised in the Show Cause Order, MISO modified its proposal to add the market registration requirement for ZG Resources greater than 5 MW that have a ZG Associated Load. This change reflects appropriate cost causation principles, and will ensure that Large Loads are responsible for market payments

based on their actual output, and not netted based on the presence of co-located generation. Additionally, the ZG Associated Load will under existing Tariff rules be required to obtain Transmission Service. MISO has not expanded the scope of ZG Interconnection Service to address "proximate location" service and related matters raised in the Show Cause Order because the scope of ZG Interconnection Service is core part of its design. However, ZGIS does not preclude MISO from developing other process improvements. MISO has begun work on further filings, including those that address transmission service refinements relating to large load, that will complement ZGIS and address matters discussed in the Show Cause Order.

VIII. CONCLUSION

Q. IN YOUR OPINION, IS MISO'S FILING JUST AND REASONABLE?

A. Yes. It addresses certain specific needs that MISO, the Commission and the stakeholder process identified. The proposed new Tariff revisions address noted Commission and stakeholder concerns and will significantly enhance MISO's ability to ensure resource adequacy and reliability needs. The proposed Tariff revisions will enable a pathway for certain large load and generation configurations to achieve speed to reliable power and support the responsible growth and use of these assets, while offering customers optimized transmission solutions and greater certainty to support siting, financing, and state coordination.

Q. DOES THIS CONCLUDE YOUR TESTIMONY?

A. Yes, it does.

Affidavit of Marc Keyser

COUNTY OF HAMILTON)
)
STATE OF INDIANA)

Marc Keyser, being duly sworn, deposes and states that he prepared the Prepared Direct Testimony of Marc Keyser, and the statements contained therein are true and correct to the best of his knowledge and belief.

Marc Keyser

SUBSCRIBED AND SWORN BEFORE ME, this 12th day of August, 2026.

JENNIFER L MILLER
Notary Public, State of Indiana
Boone County
Commission Number NP0697079
My Commission Expires
February 22, 2033

Attachment A

Attachment A

Table of Stakeholder Meetings and Materials

MEETING DATE	MATERIALS
December 3, 2025 PSC	Consideration of Zero-Injection GIA
January 21, 2026 PAC	Consideration of Zero-Injection GIA
February 25, 2026 PAC	Market Participation and Registration of Co-Located Load and Generation (PAC-2024-4): Consideration of Zero-Injection GIA
April 20, 2026 LLWG	Large Load Overview Consideration of Zero-Injection GIA Resource: Adequacy Participation and Markets and Reliable Operations
April 22, 2026 PAC	Consideration of Zero-Injection GIA (Interconnection and Planning Draft Tariff Updates): Market Participation and Registration of Co-Located Load and Generation (PAC-2024-4A)
May 14, 2026 LLWG	Consideration of Zero-Injection GIA (Resource Adequacy, Markets & Reliable Operations Draft Tariff): Market Participation and Registration of Co-Located Load and Generation (PAC-2024-4B) ZGIA Q&A
May 27, 2026 PAC	Consideration of Zero-Injection GIA (Interconnection and Planning Draft Tariff Updates): Market Participation and Registration of Co-Located Load and Generation (PAC-2024-4A)
June 10, 2026 LLWG	Consideration of Zero-Injection GIA: Market Participation and Registration of Co-Located Load and Generation (PAC-2024-4B)
June 29, 2026 LLWG	Consideration of Zero-Injection GIA: Market Participation and Registration of Co-Located Load and Generation (PAC-2024-4B)
July 15, 2026 PAC	Liaison Report
July 16, 2026 LLWG	MISO Posted Items: ZGIA Market Participation and Registration of Co-Located Load and Generation: Consideration of Zero-Injection GIA (PAC-2024-4B)
July 27, 2026 LLWG	Liaison Report
August 12, 2026 LLWG	Posted Tariff revisions